



From Traditional Web to AI-driven Web

Outline

I. World wide web

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- Information retrieval
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- AI-free content

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What is the World Wide Web?

World Wide Web (WWW)

- The leading information retrieval service on the Internet
- Internet : Global network connecting computers
- Enables access to information through web browsers (e.g., Chrome, Firefox, Edge)

Web Content

- Any content accessible online, including: Text (TXT), Images (JPG, PNG), Audio (MP3), Video (MP4), Data sets (CSV)
- Typically created using HTML and hosted on websites

Key facts:

- The web hosts over 1.5 billion websites, though only about 200 million are active.
- Around 67% of the global population has internet access
- English is the dominant language of online content (around 50%)

A website

- A collection of interconnected web pages
- Hosted under a single domain name
- Accessed through a web address (URL)

Web Address

URL = *Uniform Resource Locator*

unique web address of an online resource

<http://www.balamand.edu.lb/aboutuob>

- **Protocol:** http:// or https:// (secure)
- **Subdomain (optional):** e.g., www.
- **Domain name:** e.g., balamand
- **Top-Level Domain (TLD):**
 - Generic TLDs: .com, .org, .edu, .gov
 - Country code : .us, .fr, .lb
- **Path to specific page :** e.g., /about-us

Exercise: Type of websites

- The objective of this exercise is to help students **identify and classify different types of medical and pharmaceutical websites**. It also develops their ability to **analyze website purpose, content, and target audience**.
- Search query: health and pharmaceutical information

Types of websites

- Government sites
- Educational sites
- News and media
- NGO websites
- Company websites - E-commerce (Amazon, eBay) –
- Blogs - Social networks (Facebook, X, Instagram)
- Forums (Reddit, Quora) - Streaming platforms (YouTube, Netflix)
- Data platforms (data.gov)
- Open access scholarly Research.



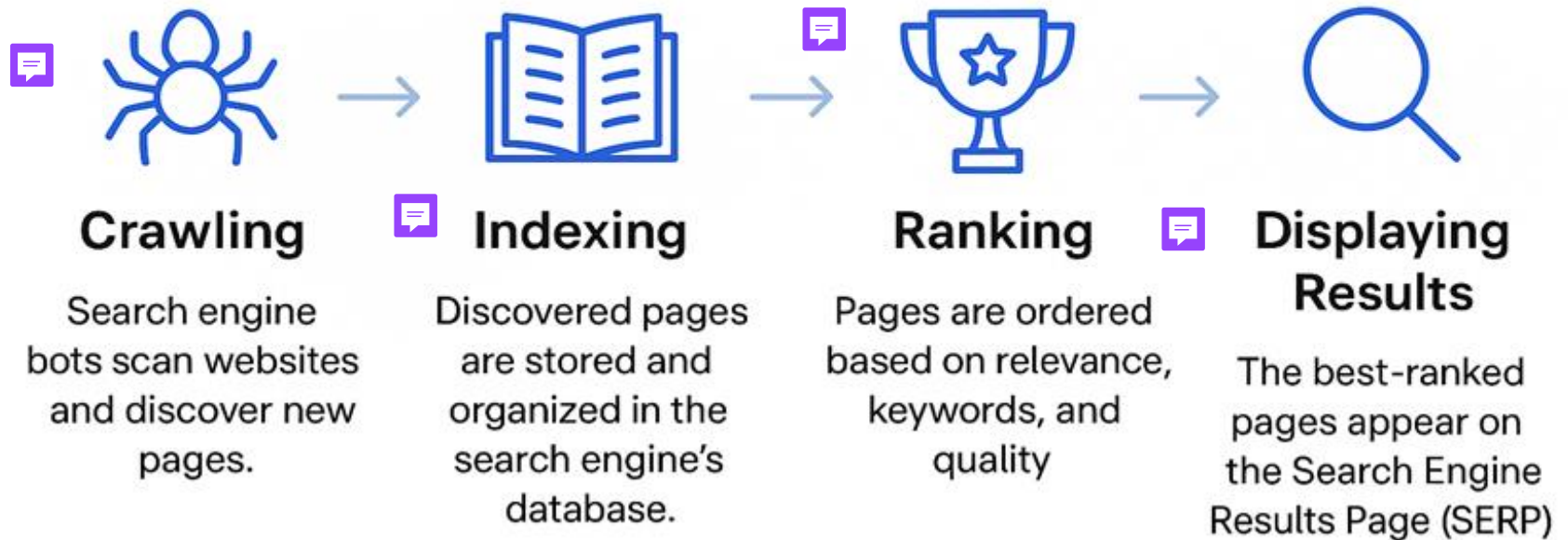
- Open educational resources (**MOOC**: Massive Open Online Course <https://www.edx.org/> , Coursera, EdX, Khan Academy)

Layers of the web

Information on the web can be broadly divided into **three layers**:

Layer	Description	Access
Surface Web	Publicly accessible web pages indexed by search engines (news, blogs, open sites).	Open & Free access
Deep Web	Databases, academic repositories, library catalogs, intranets — not indexed by standard search engines.	Requires login or query
Dark Web	Encrypted networks (e.g., TOR) used for privacy or illicit activities.	Restricted / anonymous

How Websites Appear on Search Engines



Ranking: they employ algorithms to display results based on relevance ([page rank](#))

Trafficking: refer to the volume and activity of users to a certain website or webpage ([web traffic](#))

 **Search engine optimization (SEO)** is the practice of improving a website's visibility in unpaid search engine results to increase the quantity and quality of its traffic.

Google Family

- **Google Books** is a service that searches the full text of books that Google has scanned, and stored in its digital database
<http://books.google.com>
- **Google Scholar** is a product that searches for scholarly literature and academic resources from journals, theses, and books. The ranking of results is by the number of citations of each item
<http://scholar.google.com>
- **Google products, services, and Apps.:** Translate, News, Trends, Images, Earth, Maps, Videos, Play, Travel, ...
- **Google Advanced Search**

Search by: All words, Exact phrase or quote, Any or None of the words

Narrow/refine/filter by: Language, Region, Date of last update, Site or domain (.edu / .org / .gov / .com) ,File type (Word / Excel / PDF / PowerPoint / ...)

https://www.google.com/advanced_search

AI Golden Prompt

- A **golden prompt** highly structured, specific AI instructions designed to produce high-quality, consistent, and relevant outputs.
- **Role:** "Act as an experienced instructional designer...",
- **task :** "Summarize," "create," "explain," "generate," "analyze"
- **Context,** : "...this article on climate change..."
- **Output format** "...in 300 words, focusing only on key takeaways in bullet points".
-  In short: **Be clear, specific, and tell the AI what you want, how you want it, and why.**

Prompt 1 :

Explain Photosynthesis.

Prompt 2:

Role: Act as a Senior-Level Biology Student specializing in plant physiology and bioenergetics.

Context: I am preparing for a comprehensive exam and need a high-level refresher on the biochemical pathways of energy conversion. Assume I already understand basic plant anatomy.

Goal: Explain the process of Photosynthesis, specifically focusing on the coupling of the Light-Dependent Reactions (Photosystems I and II) with the Calvin Cycle. Detail the role of the electron transport chain (ETC) and the generation of the proton motive force.

Output: Provide a structured technical summary. Use bullet points for the step-by-step molecular flow and a concluding table that summarizes the net inputs and outputs for one full cycle. Keep the tone academic, precise, and professional.

Prompt

Create a vertical comparison infographic titled 'How Media Drives Purchases' using a professional flat-vector style.

Data to include: Online Reviews (93%): Top tier, labeled 'The Trust Factor.' Social Media Ads (74%): Labeled 'The Engagement Driver.' TV Advertising (53%): Labeled 'Brand Awareness.' Online Banners (31%): Labeled 'Regional Influence.' Print Media (2%): Labeled 'Niche Reach.'

Design Specs: Use a 'Power Ranking' layout with bars of descending length. Use a color gradient from deep teal (high influence) to light gray (low influence). Keep text minimal, legible, and use simple icons for each channel. Aspect ratio 9:16 for mobile viewing."

Output

How Media Drives Purchases



93% Online Reviews
The Trust Factor



74% Social Media Ads
The Engagement Driver



53% TV Advertising
Brand Awareness



31% Online Banners
Regional Influence



2% Print Media
Niche Reach

Evaluate online content

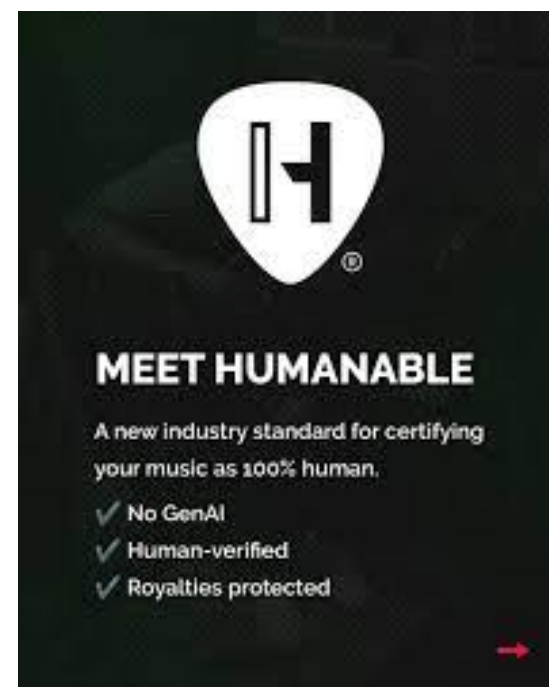
A crucial task to ensure that the information is accurate, reliable, and trustworthy

Consider the source:

- **Authority:**
 - Who is responsible for the info?
 - Who wrote it?
 - Who published it online?
 - Who developed the site?
 - Always check the domain (.edu, .gov, .org, ...)
 - Chat GPT Lacks Clear authorship.
 - Look for bias.
- **Timeliness:**
 - How current and up-to-date is the published information?
 - When was the page created or last updated?
- **Objectivity:** Try to distinguish between promoting and serious content. Beware of bias sites!

FAKE information

- ❑ Sharing misinformation can be **intentional** or **nonintentional**
- ❑ **Don't share information unless you check its credibility!!**
- ❑ **AI Deepfakes** are images, videos, or audio that have been edited or generated using artificial intelligence, AI-based tools or audio-video editing software.



AI-Free™ Certification

Purely Human, Truly Certified.

The AI-Free™ certification allows you to proudly display your creative work as free from the influence of artificial intelligence.

In a time of AI-generated content, your certification badge serves as a stamp of authenticity, highlighting the human creativity and craftsmanship that went into your artistic work.

[Get Certified Now →](#)

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Online & AI Privacy Tips



Secure devices: Use strong passwords and 2FA



Private search: Use Startpage or DuckDuckGo



Incognito mode: Browse without saving history



Check permissions: Limit app access



Update regularly: Block cookies & trackers



Think before sharing: Review social media privacy



Avoid public Wi-Fi: Don't access sensitive info



Clear cookies: Remove trackers often



Encrypted chats: Use secure messaging apps



Stay informed: Learn about AI privacy risks

Cyber Ethics

- ❑ Refers to the basic ethics and etiquette that must be followed while the internet

- ❑ **Cyber responsible behavior includes:**
 - Respecting privacy and avoid unauthorized access
 - Securing personal information
 - Avoiding cyberbullying and harassment
 - Verifying information
 - Respecting intellectual property

[Practice Cyberethics](#)